MIL-C-48514(PA)
30 September 1974
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

CASE SECTION, BALLISTIC, CENTER, AFT:
M72, M86, M87

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of shell and frame assembly entitled as Shell and Frame Assembly, Middle Section, M72, M86 and M87 which are used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-60	- Tape, Pressure Sensitive Adhesive Waterproof for Packing and Packaging.

MILITARY

MIL-S-48512	- Shipping and Storage Container, Atomic Warhead Section: M480E1 and M480.
MIL-T-5021	- Test, Aircraft and Missile Welding Operators Qualification.
MIL-W-8604	- Welding of Aluminum Alloys Process for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

FSC1115

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

DRAWINGS

ORDNANCE CORPS

8822030	- Box, Packing, Ammunition, for Case Section Ballistic, Aft.
8824386	- Shipping, Container, Case, Atomic Warhead Section.
8825083	- Shell and Frame Assembly, Middle Section M72, M86 and M87.
8825095	- Skin, Middle Section.
8830816	- Adaption Kit, Atomic Warhead to Rocket Training M87.
8830832	- Warhead Section, Atomic, Training M72.
8831129	- Mark 31, Warhead Compatability.
8839543	- Box, Packing, Ammunition, for Case Section Ballistic, Aft.
8839593	- Ammunition Repair Parts for Rocket Motor.
EL-8842561	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting offecer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The assembly shall comply with all requirements specified on the drawing (dwg) 8825083, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-N-A. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall not be returned).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

3.5 Mating assembly.

3.5.1 Adaption kit assembly. The Shell and Frame Assembly, Middle Section, shall be capable of mating mechanically when attached to mating assemblies shown in dwg 8830823 or their equivalents when tested as specified in 4.4.1.

3.6 Warhead compatibility. The Shell and Frame Assembly, Middle Section, shall be capable of mating mechanically with the applicable mating assemblies shown on dwg 8831129 or their equivalents. The test shall be conducted as specified in 4.4.2 (M86 only).

3.7 Environmental testing. The assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The assembly shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.7.1 Waterproofness. The assembly shall allow no leakage inside the assembly when subjected to water spray simulating heavy rain for a period of 15 minutes when tested as specified in 4.4.3.1 (M86 only).

3.7.2 Temperature shock. The assembly shall withstand temperature shocks between minus (-)65 plus or minus (±)5 degrees Fahrenheit (°F) and plus (+) 160 ± 5°F without evidence of damage to protective finishes, or permanent warpage, and shall comply with 3.5.1 after being tested as set forth in 4.4.3.2.

3.7.3 Handling shock and vibration. The assembly when mounted as part of the 8830823 assembly or mounted to the equivalent in weight, balance and mounting surfaces shall withstand, without damage, the applicable requirements of the shock test of MIL-S-48512(PA) when tested as specified in 4.4.3.3. Upon completion of this test, the assembly shall comply with the requirements of 3.5.1 and 3.6. (M86 only).

3.8 Deterioration. Any internal or external components which show evidence of deterioration or corrosion shall be considered defective and the assembly removed from the lot. (M86 only).

3.9 Welding. Welded joints shall be reasonably smooth and free from craters. In addition, they shall exhibit characteristics of fusion; penetration; soundness of welded joints, representative of good welding practices. All welding fluxes, scale, weld spatter, acids or basic solutions shall be removed completely prior to application of finish coat. Welding shall be accomplished in accordance with MIL-W-8604 and shall be performed only by welding operators who have met the requirements of MIL-T-5021 for the particular alloys and types of welding involved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

3.10 Paint adhesion. The painted surface of the assembly shall adhere to the base metal without any evidence of the removal of paint exposing any coating or base metal when tested as specified in 4.4.4.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8825083) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Mating with adaption kit (see 3.5.1)		X
3. Waterproofness (see 3.7.1)		X
4. Handling shock and vibration (see 3.7.3)	X	
5. Temperature shock (see 3.7.2)	X	
6. Paint adhesion (see 3.10)		X
7. Warhead compatability (see 3.6)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Skin, Middle Section (see dwg 8825095).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect diameter at station 100.0 and 32.7	Gage
	102. Improper skin contour	Gage
	103. Inaccurate length	Gage
	104. Improper welding (Dye penetrant required)	Visual/Manual
	105. Insufficient paint coverage (applicable when assembly is procured as a separate item)	Visual
	106. Incorrect weldments	Gage
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.2 Shell and Frame Assembly, Middle Section (see dwg 8825083).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Concentricity at station 100.0 and 32.7 incorrect	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

2. Perpendicularity of forward face at station 32.5 with functional centerline incorrect.....Gage

Major:

101. Inaccurate location of 4 slots and notches at station 32.50.....Gage
102. Inaccurate location of 4 holes at station 32.50.....Gage
103. Incorrect size of 32 holes and size of countersink at station 100.0.....Gage
104. Protective finish missing or bare spots (interior and exterior).....Gage
105. Parts missing, damaged or loose.....Manual/Visual
106. Faulty function of door hinges and latch.....Manual
107. Objective evidence relating to material missing or improper (see 3.1).....Visual
108. Serialization improper or missing (see 3.4).....Visual

Minor:

201. Dents in excess of 0.05 inch maximum depth, concave with a maximum diameter of 2 inches, no limitation of spacing.....Gage
202. Scratches - 0.010 to 0.030 inch depth shall not exceed 10 inches in length - scratches of less than 0.010 inch in depth, no limitation in length and width.....Gage
203. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.3 Box, packing, ammunition, for Shell and Frame Assembly, Middle Section prior to closing (see dwg 8839543).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Missing or improper pads	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

102. Connector not wrapped or improperly wrapped.....Visual
103. Missing humidity indicator card.....Visual
104. Incorrect position of case section.....Visual
105. Missing or mislocated desiccant bag.....Visual
106. Excess air in desiccant bag.....Visual
107. Cable ends not tied securely.....Visual
108. Missing, loose or mislocated cleats.....Visual
109. Missing, loose or mislocated supports.....Visual
110. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.4 Box, packing, ammunition, for Shell and Middle Section (see dwg. 8839543).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Board, broken, cracked or split	Visual
	102. Missing or incorrect skid	Visual
	103. Missing or incorrect marking	Visual
	104. Missing, loose, mislocated or incorrect strapping	Visual
	105. Evidence of poor workmanship (see 3.11)	Visual
Minor:	None defined	

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. Handling shock and vibration	Major
b. Paint adhesion	Major

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Handling shock and vibration. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.3.3. If any item fails to comply with 3.7.3, the lot shall be rejected.

Table IV

Sample sizes for non-destructive testing

Lot Size	Sample Size
2-180	2
181-500	3
501-800	5
801-1300	7

4.3.3.2 Paint adhesion. Each assembly in the lot shall be tested in accordance with 4.4.4. If any item fails to comply with 3.10, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8842561). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

4.4.1 Mating with adaption kit assembly. The Shell and Frame Assembly, Middle Section shall be mounted to mating assemblies or a fixture equivalent to the mating surfaces on dwg 8830823.

4.4.2 Warhead compatibility. When testing for Warhead assembly requirements, the functional cargo, described in dwg 8831129 Mark 31 Warhead Compatibility information or inert equivalent shall be used. The functional cargo shall be capable of being assembled with the case section, ballistic, center, aft without distortion or damage of any parts and without the use of excessive force. (M-86 only).

4.4.3 Environmental testing (see 3.7).

4.4.3.1 Waterproofness. The assembly shall be mounted horizontally and impacted with a water spray directed within 15° of vertical. The rain test chamber temperature shall be maintained at 80 ± 20°F throughout the test period. A simulated rain fall of 10 ± 2 inches per hour shall be produced by means of a water spray nozzle. The rainfall shall be dispersed uniformly over the test area. During the test period, the assembly shall be rotated once about the longitudinal axes. Following the exposure period, the assembly shall be removed from the test chamber, wiped dry externally, disassembled from the rest of the 8830823 assembly and checked for water leakage. (M-86 only).

4.4.3.2 Temperature shock. The assembly shall be placed within a test chamber where the temperature is -65 ± 5°F. The assembly shall be subjected to this temperature for a period of 5 hours minimum, at the conclusion of which, and within 5 minutes, the assembly shall be transferred to a chamber having an internal temperature of 160 ± 5°F. The assembly shall be subjected to this temperature for a period of 4 hours minimum. This completes one cycle. The number of complete cycles shall be three. At the conclusion of the third cycle, the assembly shall be removed from the test chamber and within a period of 30 minutes shall be tested as specified in 4.4.1.

4.4.4 Paint adhesion. This test shall be performed on each assembly, after the paint has been permitted to dry for a minimum of 24 hours. Adhesive tape 3/4 inch width, conforming to PPP-T-60, Type II or III Grade A or B shall be used for this test. Firmly press a two inch length of longer piece of tape onto the painted surface of the assembly. Press out air bubbles underneath tape. Allow 10 seconds for the assembly to return to room temperature. Firmly grip a free end of the tape. Rapidly strip the tape from the surface by pulling the tape back to back upon itself - 180 degrees. Closely observe for bare spots around and under tape area. Disregard flecks of paint on tape. However, removal of paint exposing any coating or base metal shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be packaged in accordance with dwg 8822030 using the specified moisture barrier.

5.1.2 Level C. The assembly shall be packaged in accordance with dwg 8822030 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The assembly shall be packed in accordance with dwg 8822030.

5.3 Marking. Marking shall be as specified in accordance with dwg 8822030.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2403 dated 22 September 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A047

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:30Z
Check the source to verify that this is the current version before use.